Exploratory Legal Agent Evaluation: Base `LAB/OpenCode` vs. `ClaimLayer / Deltoryx` Augmented Workflow on Harvey Legal Agent Benchmark (LAB)

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Platform Checkout Inspected: Commit `d70bf3ffb19205984e6b4667aa3ad80b0e5831f8`
Benchmark & Harness Context: harveyai/harvey-labs (`b58a28a`) and google-research/harness2 (`arXiv:2610.14224`)
Agent Substrate: `OpenCode` (`v1.14.18`, `tool_mode="base"`, `skill_mode="native"`)
Solver Model: `openai/gpt-4.1`
Evaluators (Paired Within Each Task): `harvey-labs/evaluation/run_eval.py` using `openai/gpt-4.1` (Litigation, Banking) and `nvidia/nemotron-3-super-120b-a12b` (Structured Finance, Corporate M&A)
Scope: 4 selected multi-document legal tasks, 20 source documents, 199 rubric evaluation criteria

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1. Study Design and Boundaries

This report summarizes the artifact-based reassessment of an exploratory integration comparing a base `LAB/OpenCode` agent against an augmented workflow incorporating a **ClaimLayer / Deltoryx** component tree (`c1`–`c8`: system prompts, guardrails, memory, tool guidance, `c5: skills/evidence_intelligence/SKILL.md`, and `c7: scripts/deltoryx_audit.py`).

1.1 Experimental Configuration

- **Solver & Execution Limits:** Both arms use `openai/gpt-4.1` with `tool_mode="base"`, `skill_mode="native"`, 80 maximum turns, one continuation, and a default 900-second timeout.
- **Prompt & Workflow Differences:** Saved metrics record `2,333` always-on prompt characters for the base arm and `9,456` for the augmented arm, along with differing `opencode_only` flags. The augmented workflow combines document-to-text mirrors, a local SQLite claim store, source-tier labels, clause-difference tables, instructions to index and verify, and a deliverable container check.
- **Evaluation Scope:** Four tasks were selected as a convenience sample covering litigation, structured finance, corporate M&A, and banking (`34`, `37`, `62`, and `66` rubric criteria respectively; `199` correlated criteria across 4 cases). This is an author-run exploratory study using the LAB evaluator, not an evaluation or endorsement by the benchmark or `Harness²` authors, nor a reproduction of `Harness²` recursive optimization.

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2. Paired Results After Document-Container Repair

Table 1 reports the recalculated descriptive results from paired saved `scores.json` evaluations where **both arms within each task share the same judge** after document-container repair. We report **criterion pass rate** (not factual accuracy or complete-task pass rate; under the evaluator's all-criteria rule, complete-task success is `0/4` in both arms).

2.1 Paired Task-Level Breakdown

Task (practice-area / slug)	Docs	Criteria	Base Passed	Augmented Passed	Difference (pp)	Criterion Transitions (Gain / Loss)	Judge
litigation-dispute-resolution analyze-counterparty-motion-to-dismiss	6	34	1 / 34 (2.94%)	**19 / 34 (55.88%)**	**+52.94**	+18 / −0	openai/gpt-4.1
structured-finance-securitization compare-offering-memorandum-against-indenture	3	37	3 / 37 (8.11%)	**12 / 37 (32.43%)**	**+24.32**	+10 / −1	nvidia/nemotron-3-super-120b-a12b
corporate-ma analyze-counterparty-spa-markup	6	62	15 / 62 (24.19%)	**24 / 62 (38.71%)**	**+14.52**	+12 / −3	nvidia/nemotron-3-super-120b-a12b
banking-finance analyze-counterparty-markup-of-senior-secured-credit-facility-term-sheet	5	66	43 / 66 (65.15%)	**44 / 66 (66.67%)**	**+1.52**	+13 / −12	openai/gpt-4.1
Total (Pooled)	**20**	**199**	**62 / 199 (31.16%)**	**99 / 199 (49.75%)**	**+18.59**	**+53 / −16**	**Mixed (Matched per task)**

2.2 Summary Metrics

Measure	Base Arm	Augmented Arm	Difference / Ratio
Pooled Criteria Passed	62 / 199 (31.16%)	99 / 199 (49.75%)	**+37 criteria (+18.59 pp, 1.60x)**
Unweighted Mean of Task Pass Rates	25.10%	48.42%	**+23.32 pp**
Criterion-Level Gains / Losses	—	+53 gained / −16 lost	Net +37 criteria
Oracle Criterion Union (Post-hoc across both arms)	—	115 / 199 (57.79%)	Banking union: 56 / 66 (84.85%)

Measure	Base Arm	Augmented Arm	Difference / Ratio
Oracle Whole-Deliverable Selector (1 response/task)	—	`99 / 199` (`49.75%`)	Selects augmented arm on all 4 tasks
Recorded Solver Tokens	`220,320`	`388,406`	**`+168,086` tokens (`+76.3%`)**
Logged Solver Cost (Historical USD, excl. judge)	`USD 1.8000`	`USD 2.4705`	**`+USD 0.6705` (`+37.3%`)**

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3. Methodological Notes: Artifact Repair, Judge Sensitivity, and Oracle Union

3.1 Why Earlier Unpaired "Strict" Aggregates (`47/199` vs. `106/199`) Were Replaced

An earlier draft reported `47/199` (`23.62%`) versus `106/199` (`53.27%`). Reassessment of the saved evaluation files identified three reasons why that aggregate is not retained as a primary controlled comparison: 1. **Pre-Repair Container Failures in Corporate M&A:** Saved pre-repair files show `0/62` for *both* corporate M&A arms before container repair (when raw Markdown was written to `.docx` filenames, causing `pandoc` ZIP unpacking to fail), not only for the base arm. 2. **Judge Differences on Repaired M&A:** After container repair, the augmented M&A deliverable scored `31/62` under `openai/gpt-4.1` and `24/62` under `nvidia/nemotron-3-super-120b-a12b`, while the repaired base deliverable was scored under Nemotron (`15/62`). 3. **Structured Finance Readability Stage:** An earlier `openai/gpt-4.1` evaluation on structured finance scored `0/37` due to an artifact read failure, whereas the later Nemotron evaluation scored `12/37` (vs. `3/37` for the base arm under Nemotron).

The `106/199` aggregate combined the `31/62` GPT-4.1 M&A score with the later structured-finance score while retaining the `0/62` un-repaired base M&A score (`47/199`). Table 1 above instead enforces **matching judges within each task** and evaluates both repaired M&A containers (`15/62` vs. `24/62` under Nemotron), yielding `62/199` (`31.16%`) versus `99/199` (`49.75%`).

3.2 Oracle Criterion Union vs. Realizable $k = 2$ Selection

Across the Table 1 evaluations, the post-hoc union of criteria passed by either the base arm or the augmented arm is **`115 / 199` (`57.79%`)**, and **`56 / 66` (`84.85%`)** in *Banking & Finance*. - **Why this is not a $k = 2$ selector score:** This union uses ground-truth rubric verdicts to count any criterion passed in either output. A realizable selector that chooses one complete deliverable per task cannot combine mutually exclusive per-criterion advantages from two separate documents. Even an oracle whole-response selector would pick the augmented deliverable in all four tasks and remain at **`99 / 199` (`49.75%`)** (`44 / 66` in Banking). - **Implication for Agent Harnesses:** The `+53` gains and `−16` losses (`+13 / −12` in Banking) demonstrate substantial criterion-level complementarity between a playbook-focused pass and a full-text diff pass, motivating **rubric-blind candidate synthesis/merging** rather than winner-take-all rollout selection. - **No Direct Comparison to Published `Harness²` Table 2:** Published `Harness²` scores (e.g., `81.9%` at $k = 1$ and `84.5%` at $k = 5$ with Gemini 3.5 Flash) use different foundation models, task subsets, and recursive training/selection protocols, and should not be directly equated with the single-task Banking oracle union.

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4. Qualitative Observations and Limits of Attribution

4.1 Qualitative Task Patterns

- **Litigation (`1/34 → 19/34, +18 / −0`):** Pre-indexing plain-text mirrors of the 6 court filings allowed the solver to complete the opposition brief within the turn budget and address FRCP Rule 9(b) particularity, forum-selection scope, and accrual timing criteria.
- **Structured Finance (`3/37 → 12/37, +10 / −1`):** Document indexing and hierarchy prompts helped surface discrepancies between the Indenture and the Offering Memorandum (such as Reserve Account replenishment priority and Class C Note coupon differences).
- **Corporate M&A (`15/62 → 24/62, +12 / −3`):** Extracting tabular schedules alongside SPA text helped connect working-capital collar modifications to the balance sheet figures in `dataforge-financial-summary.xlsx`.
- **Banking & Finance (`43/66 → 44/66, +13 / −12`):** Full-text markup diffing recovered 13 criteria where the negotiation playbook was silent (such as deleted maturity extension options), but lost 12 criteria passed by the base run, yielding a net gain of 1 criterion.

4.2 Bundled Intervention and Implementation Boundaries

1. **Bundled Intervention:** Because plain-text mirrors, clause-difference tables, longer system/skill prompts (`9,456` vs. `2,333` chars), increased solver token expenditure (`+76.3%`), and container checks were introduced together, the `+18.59 pp` improvement cannot be causally attributed to a single mechanism without component ablations and a parsing-matched baseline.
2. **Differences Between the Inspected Audit Script and the Formal v2 Specification:**
    - `benchmarks/deltoryx_lab/scripts/deltoryx_audit.py` assigns document authority via filename heuristics, extracts at most the first 80 paragraphs per document with page locators set to `1`, and assigns fixed confidence constants by source tier.
    - Records are labelled `deductive` without executing the formal rule-and-premise proof checker specified in Section 6 of the paper, and opinion values use fixed constants rather than empirical calibration or dependence-aware scoring.
    - The `verified=true` status in the 4 saved `DELTORYX-MERKLE-SHA256-v2` certificates represents a cryptographic self-check over recorded artifacts, not a proof of semantic accuracy or complete rubric satisfaction.

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5. Repository Artifacts

- **Component Tree:** `benchmarks/deltoryx_lab/`
- **CLI Indexing & Verification Script:** `benchmarks/deltoryx_lab/scripts/deltoryx_audit.py`
- **Skill Protocol:** `benchmarks/deltoryx_lab/skills/evidence_intelligence/SKILL.md`
- **A/B Benchmark Runner:** `benchmarks/run_ab_benchmark.py`